EXPERIMENT 16

High Level Programming Extensions – Functions

Aim

To create and execute a user-defined MySQL function that calculates the annual salary from the monthly salary.

Software Required

MySQL Server and MySQL Workbench or MySQL command-line client.

SQL program:

```
CREATE DATABASE IF NOT EXISTS CompanyFunctionDB;
USE CompanyFunctionDB;

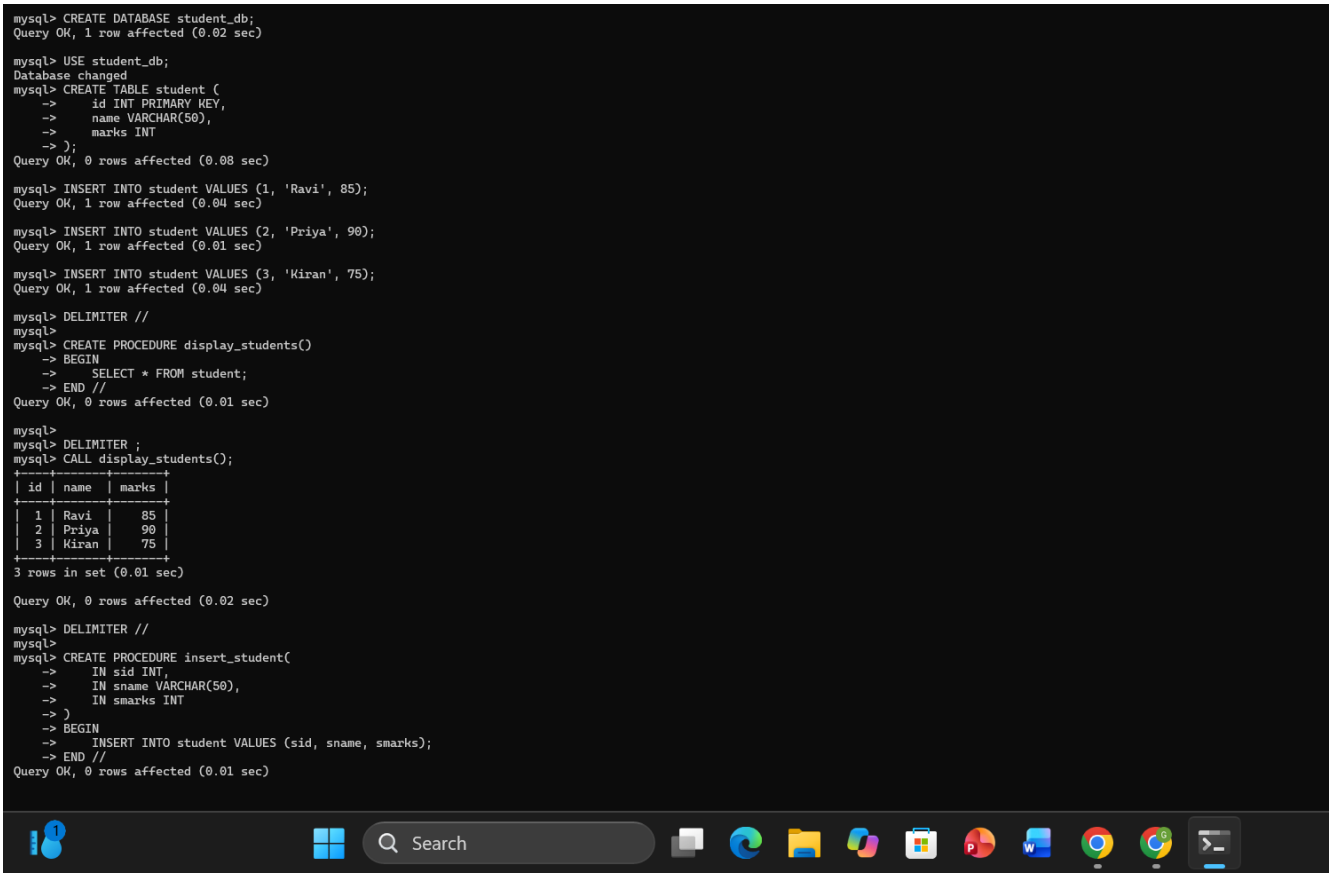
DROP TABLE IF EXISTS Employee;
CREATE TABLE Employee (
   emp_id INT PRIMARY KEY,
   emp_name VARCHAR(50),
   monthly_salary DECIMAL(10,2)
);

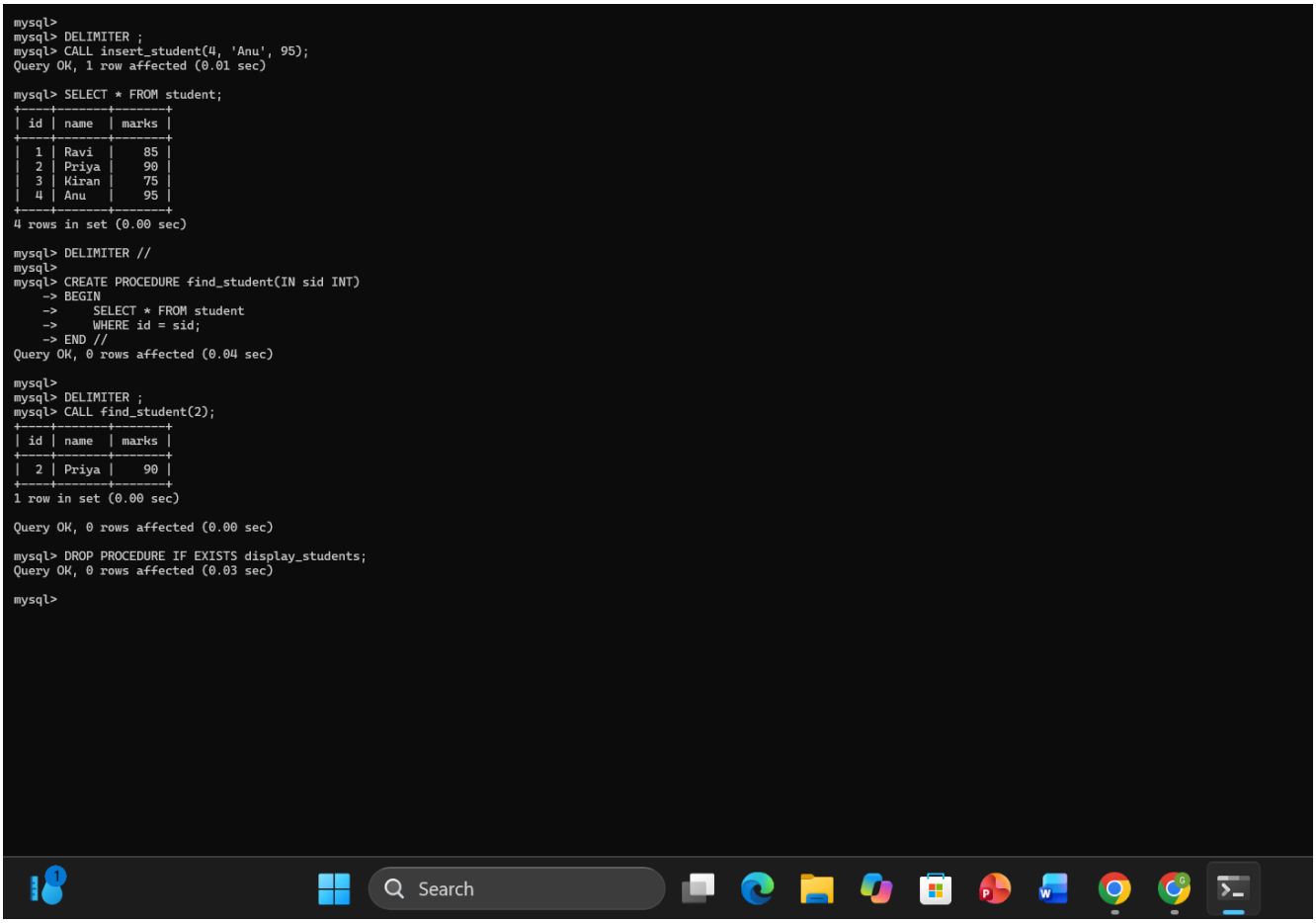
INSERT INTO Employee VALUES
(201, 'Kiran', 30000.00),
(202, 'Anu', 42000.00),
(203, 'Sanjay', 50000.00);

DROP FUNCTION IF EXISTS AnnualSalary;
DELIMITER //
CREATE FUNCTION AnnualSalary(monthly DECIMAL(10,2))
RETURNS DECIMAL(12,2)
DETERMINISTIC
BEGIN
   RETURN monthly * 12;
END //
DELIMITER ;

SELECT emp_id, emp_name, monthly_salary,
      AnnualSalary(monthly_salary) AS annual_salary
FROM Employee;
```

Expected Output:





Observation

The user-defined function accepts a monthly salary and returns the annual salary by multiplying it by 12.

Result

Thus, a MySQL user-defined function was created and executed successfully.